

Projective Reed–Solomon syndromes beyond redundancy four: deep holes, coherent polar flags, and modular carriers

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July 2026

Abstract

A projective Reed–Solomon syndrome determines a Hankel linear system of binary forms. We call the direction split-free when that system contains no completely split squarefree member. This reformulation gives a complete classification of the redundancy-five deep-hole directions over every prime power $q \geq 7$; the generic case is controlled by an integral off-diagonal fiber-square curve of arithmetic genus one.

For higher redundancy we introduce coherent polar flags: iterated contractions in which every selected factor remains a marked forbidden root. After separating the level-specific contained-component problem from the transverse point-count argument, this method gives complete all-field classifications at redundancies six and seven; the redundancy-seven split-free list is a deep-hole classification whenever the covering radius is six, in particular for $q \geq 11$. The three-marker argument at redundancy eight leaves only the persistent families for $q \geq 43$. At redundancy nine the analogous conclusion for $q \geq 53$ remains conditional on the printed slice and base-selection gates.

In characteristic two we determine the geometric degeneracy strata of an ordered Hessian pullback and isolate the remaining root-compatible component and base-selection gates. We also analyze the power-of-two Lucas arithmetic and the distinguished degree-nine e_7 orbit. The remaining degree-nine carrier strata and redundancy-ten deep holes are not classified.

1 Introduction

Let $\text{PRS}(k)$ denote the projective Reed–Solomon code of length $q + 1$ obtained by evaluating homogeneous binary forms of degree $k - 1$ on $\mathbb{P}^1(\mathbb{F}_q)$. If the redundancy is

$$r = q + 1 - k,$$

a parity-check matrix has as columns the $\mathbb{F}_q$ -points of the degree- $(r - 1)$ normal rational curve

$$\mathcal{R}_{r-1} = \{(1, t, \dots, t^{r-1}) : t \in \mathbb{F}_q\} \cup \{(0, \dots, 0, 1)\} \subset \mathbb{P}^{r-1}.$$

Call a projective syndrome direction *split-free* if it is not contained in the span of $r - 2$ distinct columns. When $\rho(\text{PRS}(q + 1 - r)) = r - 1$, the split-free directions are precisely the projective syndrome directions of deep-hole cosets. The correspondence between deep holes, one-symbol MDS extensions, and uncovered points of a normal rational curve was developed in [Kai17, ZWK20]. Zhang, Wan, and Kaipa classified projective Reed–Solomon deep-hole directions through redundancy four and identified redundancy five as the next case [ZWK20]. Our first theorem gives the corresponding all-field redundancy-five classification.

Definition 1.1 (Working terminology). A *carrier* is a closed subvariety of a syndrome space on which the generic transverse splitting argument degenerates. A *persistent carrier* is a catalecticant component propagated by ordinary contraction; a *modular* or *Lucas carrier* is

Table 1: Notation used throughout the paper.

Symbol	Meaning
q	field size; all numerical thresholds are thresholds on prime powers
k	code dimension in PRS(k)
$r = q + 1 - k$	redundancy
$n = r - 1$	divided-power degree of a syndrome representative
E	a two-dimensional vector space over the current base field
$f \in \Gamma^n E$	divided-power representative of a projective syndrome
$W_f \subset \text{Sym}^{n-1} E^\vee$	Hankel/apolar witness system annihilated by f
g, δ, κ	genus, deletion degree, and point-bound correction
b, c	transverse bad-carrier and marker-collision degrees
$d = 2^s$	power-of-two Lucas degree; used only in Section 9

a component created by a positive-characteristic contraction kernel. A direction is *shallow* when it has a split squarefree witness. A pullback is *root-compatible* when the chosen contraction factor remains a marked forbidden root on every lower incidence. An *ordered-root cover* remembers an ordering of the roots over the squarefree splitting locus. A field is *code-admissible* for a displayed statement when it satisfies both that statement's field bound and its covering-radius hypothesis.

Status of this draft. This integrated version freezes the theorem spine and the computational trust boundaries. It is a research-announcement draft: the degree-specific proof expansions and immutable external supplement listed in the proof ledger remain release gates. In particular, this version is not described as a proof-complete submission.

Theorem 1.2 (Complete redundancy-five classification). Let $q \geq 7$ be a prime power. The covering radius of PRS($q-4$) is 4. Its projective deep-hole syndromes are exactly the following disjoint families:

- (i) the tangent family T, of size $q(q+1)$;
- (ii) the conjugate-secant family S, of size $q(q^2-1)/2$;
- (iii) if $\text{char } \mathbb{F}_q \neq 3$, the rational osculating-pair family O^+ when $q \equiv 2 \pmod{3}$, of size $q(q+1)/2$, or the conjugate osculating-pair family O^- when $q \equiv 1 \pmod{3}$, of size $q(q-1)/2$;
- (iv) in characteristic three, one $\text{PGL}_2(q)$ -fixed nucleus point and one wild Artin-Schreier orbit W of size $(q^2-1)/2$;
- (v) the sporadic $\text{PGL}_2(q)$ -orbits in Table 4, occurring exactly for $q \in \{7, 8, 9, 11, 13, 17, 19\}$.

There are no other deep-hole syndromes. The three corresponding nonsporadic totals are

$$\frac{q^2(q+3)}{2}, \quad \frac{q(q+1)(q+2)}{2}, \quad \frac{q^3+3q^2+q+1}{2},$$

according as $q \equiv 1 \pmod{3}$ with characteristic not three, $q \equiv 2 \pmod{3}$, or $\text{char } \mathbb{F}_q = 3$.

The main mechanism is a characteristic-free Hankel dictionary. A syndrome at redundancy r annihilates a linear system of binary forms of degree $r-2$. It is split-free if and only if this system contains no completely split squarefree form; it is a deep-hole direction when, in addition, $\rho(\text{PRS}(q+1-r)) = r-1$. At redundancy five the system is a pencil of cubics. Its induced degree-three map of projective lines is either cyclic or has geometric monodromy S_3 .

Cyclic descent gives the arithmetic families in Theorem 1.2; in the S_3 case the off-diagonal fiber square is an integral curve of arithmetic genus one, and the singular-curve Weil bound [AP95] forces a split fiber for $q \geq 23$. The certified finite computations in Section 10 handle the remaining fields.

Starting at redundancy six, the Hankel system is no longer a pencil. The useful replacement is the coherently parameterized line of contractions obtained by choosing one prospective root. Iteration produces a *polar flag*; every chosen root remains marked because a lower witness containing it would lift with multiplicity two. The pointed coherence is essential: at $q = 19$ an individual contracted fiber has no split witness avoiding its marker, but no consecutive-row polar line is contained in its orbit.

The resulting structural theorem has three outputs.

Theorem 1.3 (Classification results and explicit gates beyond five). The following statements hold.

- (a) For every prime power $q \geq 7$, the projective deep-hole directions of $\text{PRS}(q - 5)$ are exactly the persistent tangent and conjugate-secant families, the exceptional orbits listed for $q = 7, 8, 9, 11, 13$, and, when $q = 2^m$ with odd $m \geq 5$, one characteristic-two nucleus orbit.
- (b) The displayed finite domains through $q = 32$ have the certified split-free classification stated in Section 6. The same list continues over every field: for $q \geq 13$ it is the persistent families plus one central nucleus point when $q = 2^m$ and m is odd. The list is a deep-hole classification only when $\rho(\text{PRS}(q - 6)) = 6$, in particular for $q \geq 11$.
- (c) For $q \geq 43$, the projective deep-hole directions of $\text{PRS}(q - 7)$ are exactly the persistent tangent and conjugate-secant families. Conditional on the degree-eight six-slice and base-selection gates printed in Section 7, the same conclusion holds for $\text{PRS}(q - 8)$ when $q \geq 53$. In either case the asserted families have total size $q(q + 1)^2/2$.

At redundancy r , the semilinear orbits in the persistent conjugate-secant family are parameterized by

$$T/T^{r-1} \quad \text{modulo inversion and Frobenius,}$$

where $T = \{z \in \mathbb{F}_{q^2}^\times : z^{q+1} = 1\}$ is the norm-one torus and Frobenius is coefficient Frobenius. On the tangent family the unipotent coordinate transforms as $z \mapsto z + (r - 1)u$; consequently its orbit decomposition changes precisely when $\text{char } \mathbb{F}_q$ divides $r - 1$.

The three parts have different scopes. Parts (a) and (b) are unconditional all-field syndrome classifications, but part (b) becomes a deep-hole classification only when $\rho(\text{PRS}(q - 6)) = 6$. In part (c) the redundancy-eight clause is unconditional, while the redundancy-nine clause isolates its remaining mathematical gates rather than treating them as certificate inputs. Neither clause makes a classification claim below its displayed high-field threshold.

Relation to previous work. The tangent and conjugate-secant families and the coding/normal-rational curve dictionary come from [Kai17, ZWK20]; the covering-radius input in the high-rate range is due to Seroussi and Roth [SR86]. The broader arc/MDS and code-automorphism background is surveyed or established in [BL19, Dür87]. Classical apolarity, catalecticants, and binary-form rank strata are treated in [IK99, CS11]; our use of divided powers keeps the small-characteristic contraction formulas explicit. Binary-quartic orbit theory in characteristic different from two and three [KPP25], factorization statistics in linear families [CMP17], normal-rational-curve nuclei [GH00], and the general Frobenius/monodromy semantics of splitting families [Wan26] are inputs. The first two references identify deep syndromes with uncovered points of a normal rational curve and settle the

lower-redundancy families; they do not classify the redundancy-five cubic pencils. The orbit-theoretic and factorization references supply normal forms and distribution tools, but not the coherent marked contraction used here. The nucleus results locate characteristic-dependent kernels, while the present work computes which of their coherent lifts meet the PRS splitting incidence. Standard function-field, Artin–Schreier, and linearized polynomial background is taken from [Sti09, LN96]; the characteristic-two orbit and component exhaustions themselves are proved internally, not imported from the quartic-orbit reference. Thus the new assertions are separated into the redundancy-five through seven classifications, the conditional transverse induction mechanism, the fixed-level high-field applications, and the characteristic-two contained-carrier results. Claim-by-claim novelty and dependency boundaries are recorded in the electronic ledger; no database-search procedure is part of the mathematical argument.

Organization. Section 2 first gives a proof-free guide to the argument, and Section 3 gives the Hankel dictionary. Section 4 proves Theorem 1.2. Sections 5–7 develop coherent polar induction and prove Theorem 1.3. Section 8 gives the characteristic-two ordered-Hessian replacement. Section 9 treats Lucas-carrier arithmetic. Section 10 states the computational and formal trust boundary, and Section 12 records the exact open boundary.

2 Guide to the proof

This section is a proof-free guided tour. Its purpose is to install the information flow used throughout the paper before coordinates, finite exceptional tables, or formal trust boundaries intervene.

Syndrome to splitting system. A syndrome direction $[f] \in \mathbb{P}(\Gamma^{r-1}E)$ determines its Hankel annihilator $W_f \subset \text{Sym}^{r-2} E^\vee$. A product of $r - 2$ distinct rational linear factors lies in W_f exactly when f lies in the span of the corresponding normal-rational-curve columns. Thus the geometric classification begins with split-free systems. The separate covering-radius theorem is what promotes split-free directions to code deep holes; the dictionary alone does not.

The redundancy-five base cover. At redundancy five, W_f is a pencil of binary cubics. Its root incidence is a degree-three map of projective lines. The off-diagonal fiber square separates the cyclic cases, controlled by Frobenius on a deck transformation, from the S_3 case, controlled by rational points on a geometrically integral $(2, 2)$ curve. This is the bottom cover used at every later level.

Why independent lower fibers fail. At higher redundancy, choosing one lower witness forgets which factor was removed. If that lower witness contains the chosen factor, then multiplication restores it with multiplicity two rather than producing a squarefree form. The lost datum is therefore not auxiliary: it is the exact obstruction to lifting.

Coherent replacement and the fork. Choose a prospective root λ , contract f to $\iota_\lambda f$, and retain λ as a marked forbidden root. Iteration gives an ordered polar flag. Its first-polar line has two qualitatively different behaviors. If it is transverse to the lower bad carrier and collision schemes, only finitely many parameters must be removed and a point count produces an escape. If it is scheme-theoretically contained, the point-count argument says nothing: the persistent and modular components require a separate level-specific classification.

Lifting. On the transverse branch, a rational point of the lower ordered splitting cover outside the branch, diagonal, old-marker, new-marker, and self-collision divisors produces a split squarefree lower witness avoiding all markers. Multiplying by the marked factors then lifts it squarefreely to W_f . Figure 1 records this sequence and the contained/transverse fork.

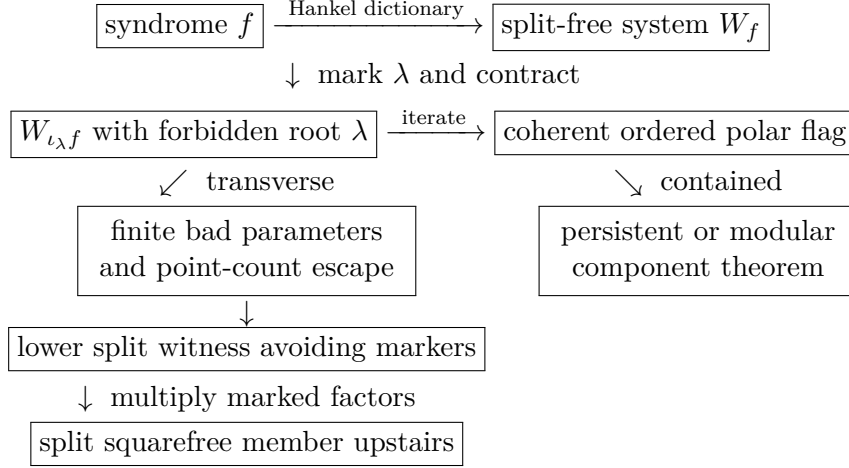


Figure 1: Information flow of coherent polar induction. Solid arrows are the statements proved in the degrees treated here. The contained branch is not a consequence of the transverse sieve.

Table 2: The five proved levels. The marker count is the number present at the bottom cubic cover.

Level	lower cover	marks	contained	conclusion
R5	cubic pencil / (2, 2) fiber square	0	tangent, sigma, cyclic families	complete all-field deep-hole classification
R6	pointed R5 cover	1	persistent plus binary nucleus line	complete all-field deep-hole classification
R7	iterated pointed cover	2	persistent plus central binary nucleus	complete all-field split-free classification; deep for $q \geq 11$
R8	three-marker R5 cover	3	persistent; modular lifts shallow	conditional on the full lower package for $q \geq 43$
R9	four-marker R5 cover	4	persistent plus checked characteristic-seven carrier	conditional persistent-only target for $q \geq 53$

The proofs now follow these arrows in order: dictionary, redundancy-five base cover, polar construction, contained components, transverse point count, and fixed-level applications. This is a scope boundary, not a second theorem inventory.

3 The syndrome–Hankel dictionary

Let E be a two-dimensional vector space. We use divided powers for syndromes and ordinary symmetric powers for root forms. This avoids invalid binomial rescalings in small characteristic. For $f = (a_0 : \cdots : a_{r-1}) \in \mathbb{P}(\Gamma^{r-1}E)$, write $e_0, \dots, e_{r-1}$ for the divided-power coordinate basis and $\nu(t) = (1, t, \dots, t^{r-1})$, with $\nu(\infty) = e_{r-1}$. The perfect divided-power/symmetric-power pairing is characterized by coefficient extraction. Define

$$H_f = \begin{pmatrix} a_0 & a_1 & \cdots & a_{r-2} \\ a_1 & a_2 & \cdots & a_{r-1} \end{pmatrix}, \quad W_f = \ker H_f \subset \text{Sym}^{r-2} E^\vee.$$

Table 3: What each passage preserves and what it does not prove.

Passage	What survives	What remains outside the passage
Hankel dictionary	split-free syndrome criterion	covering radius and code deepness
Radius promotion	split-free equals deep in stated fields	the syndrome classification itself
Transverse induction	witness outside finite bad/deletion loci	classification of contained polar lines
Contained analysis	persistent/modular components in proved degrees	an arbitrary-degree component theorem
Finite certificates	exact frozen finite domains and orbit records	geometric integrality and all-field continuation
C517 Lean synthesis	residual algebra and implication from explicit inputs	six-slice geometry, rational points, PRS identification, exhaustion, group action
Ordered Hessian	characteristic-two contained-carrier replacement	deepness of every Lucas-carrier point
Lucas arithmetic	the distinguished endpoint and e_7 orbit	remaining degree-nine strata and redundancy ten

Lemma 3.1 (Hankel criterion). Let $\lambda_1, \dots, \lambda_{r-2}$ be distinct points of $\mathbb{P}(E^\vee)(\mathbb{F}_q)$ and put $g = \prod_i \lambda_i$. Then

$$f \in \text{span}(\nu(\lambda_1), \dots, \nu(\lambda_{r-2})) \iff g \in W_f.$$

The same statement with repeated factors and Hasse derivatives gives the corresponding osculating spans. Both statements commute with $\text{PFL}_2(q)$.

Proof. On an affine chart, g is the characteristic polynomial of an order- $(r-2)$ recurrence. The length- r solutions are spanned by the geometric sequences $\nu(\lambda_i)$; their independence is the Vandermonde determinant. A root at infinity gives the reversed-chart recurrence. For repeated roots, the confluent Vandermonde basis is formed by Hasse derivatives, so the same argument works in every characteristic. Equivariance follows because substitution transports both the normal rational curve and the recurrence. $\square$

Corollary 3.2 (Split-free criterion). Assume $\rho(\text{PRS}(q+1-r)) = r-1$. A projective syndrome f is deep if and only if W_f contains no completely split squarefree form of degree $r-2$ over $\mathbb{F}_q$.

We call the latter property *split-free* even when the covering-radius hypothesis has not been established. Thus every deep syndrome in the radius- $r-1$ range is split-free, while a small-field split-free census is not silently promoted to a code deep-hole classification.

The persistent locus is already visible here. If every member of W_f has a common quadratic factor, then that factor is either a rational square, an irreducible quadratic, or a product of two distinct rational factors. These are respectively the tangent deep family, the conjugate-secant deep family, and a shallow rational secant. The common factor cannot grow with redundancy: since $\dim W_f = r-3$,

$$r-3 \leq r-1 - \deg \gcd(W_f),$$

so its degree is at most two.

4 Redundancy five: exceptional cubic covers

Here $f = (a_0 : \cdots : a_4)$ and W_f is the pencil annihilated by

$$\begin{pmatrix} a_0 & a_1 & a_2 & a_3 \\ a_1 & a_2 & a_3 & a_4 \end{pmatrix}.$$

These are divided-power coordinates; the coefficients of a cubic in W_f are ordinary symmetric-power coordinates. Factoring the formal quartic $\sum a_i T^i U^{4-i}$ would therefore be noninvariant in characteristics two and three.

Proposition 4.1 (Radius gate). For every prime power $q \geq 7$,

$$\rho(\text{PRS}(q-4)) = 4.$$

Proof. The coding/arc criterion identifies radius four with completeness of the quartic normal rational curve. Seroussi–Roth prove completeness when

$$k \geq \left\lfloor \frac{q-1}{2} \right\rfloor$$

apart from the even-field dimensions $k = 2, q-2$ [SR86]. Here $k = q-4$, the inequality holds for $q \geq 7$, and neither exceptional dimension occurs. Thus Corollary 3.2 applies. The finite certificate independently checks the same conclusion only in its declared finite domain; it is not the all-field radius proof. $\square$

4.1 Gcd strata

Proposition 4.2 (Quadratic gcd). If $\deg \gcd(W_f) = 2$, then

$$W_f = Q\langle T, U \rangle.$$

If Q is split squarefree, f is shallow. If Q is irreducible, f belongs to the conjugate-secant family; if $Q = \lambda^2$, it belongs to the tangent family. Conversely every point in those families has the displayed gcd. Their deep-family sizes are

$$|T| = q(q+1), \quad |S| = \frac{q(q^2-1)}{2}.$$

Proof. The equality follows by dimension. A split Q and a linear cofactor avoiding its two roots give a split squarefree cubic. An irreducible Q survives in every member, while a square Q forces a repeated root in every member. Lemma 3.1, including its confluent and conjugate-root forms, identifies the corresponding syndromes with the secant and tangent spans. The sizes are the established tangent/conjugate-secant counts from the lower-redundancy classification [Kai17, ZWK20]. $\square$

Proposition 4.3 (Linear gcd). Suppose $\deg \gcd(W_f) = 1$. Then f is shallow for $q \geq 8$. At $q = 7$ the same conclusion is part of Certificate R5.

Proof. Write $W_f = \lambda_0 V$ with V a basepoint-free quadratic pencil. If its degree-two map $\psi : \mathbb{P}^1 \rightarrow \mathbb{P}^1$ is separable, its rational deck involution gives $q+1$ ordered same-fiber pairs. Delete at most two fixed points, four points over branch values, and the at most two pairs whose selected member contains λ_0 . For $q \geq 8$ a surviving pair gives a split quadratic whose roots avoid λ_0 , and hence a split squarefree cubic in W_f . In characteristic two an inseparable quadratic pencil is equivalent to $\langle T^2, U^2 \rangle$; the two overlapping Hankel equations then force f onto the normal rational curve, contrary to the rank-two pencil hypothesis. $\square$

4.2 The trivial-gcd cubic cover

Choose a basis c_1, c_2 of a trivial-gcd pencil and put

$$X_f = \{([u : v], t) : uc_1(t) + vc_2(t) = 0\} \subset \mathbb{P}^1 \times \mathbb{P}^1.$$

Proposition 4.4 (Cubic incidence and off-diagonal fiber square). The curve X_f is geometrically integral of bidegree $(1, 3)$ and arithmetic genus zero. Projection to the root coordinate is birational, while projection to the pencil coordinate gives a degree-three morphism

$$\phi_f : \mathbb{P}^1 \longrightarrow \mathbb{P}^1.$$

If ϕ_f is separable, division of

$$c_1(t_1)c_2(t_2) - c_2(t_1)c_1(t_2)$$

by the diagonal bracket defines a residual curve $Y_f \subset \mathbb{P}^1 \times \mathbb{P}^1$ of bidegree $(2, 2)$ and arithmetic genus one. Its geometric components are the geometric-monodromy orbits on ordered pairs of distinct roots.

Proof. A component of X_f of bidegree $(0, e)$ would be a common factor of c_1, c_2 ; hence none exists. The bidegree genus formula gives genus zero. A root determines the unique pencil member containing it, so the root projection is birational. For a separable map the diagonal factor in the fiber-square determinant has multiplicity one. Removing its bidegree $(1, 1)$ from the fiber square leaves bidegree $(2, 2)$, whose arithmetic genus is one. The standard fiber-square/monodromy correspondence identifies its components with orbits on distinct ordered sheets. $\square$

Lemma 4.5 (Cyclic stratum). Suppose ϕ_f is geometrically cyclic.

- (i) In characteristic different from three there are two ramification points. If they are rational, the corresponding osculating-plane intersection is deep exactly when $q \equiv 2 \pmod{3}$; if they are conjugate, it is deep exactly when $q \equiv 1 \pmod{3}$.
- (ii) In characteristic three all osculating planes meet in $n = (0 : 0 : 1 : 0 : 0)$, whose pencil is $\langle T^3, U^3 \rangle$; this fixed point is deep.
- (iii) The remaining wild characteristic-three pencils have the form $\langle U^3, T^3 + aTU^2 \rangle$. They are deep exactly when $-a$ is nonsquare, forming one orbit of size $(q^2 - 1)/2$.

Proof. Assume first that the characteristic is not three. Riemann–Hurwitz gives two totally ramified points t_1, t_2 . The confluent Hankel criterion places f in $\text{Osc}(t_1) \cap \text{Osc}(t_2)$; after transporting the pair to $\{0, \infty\}$, the pencil is $\langle T^3, U^3 \rangle$ and a geometric deck generator is $t \mapsto \zeta t$.

For a rational ramification pair, Frobenius fixes the points and the deck map descends exactly when $\zeta^q = \zeta$, namely $q \equiv 1 \pmod{3}$. Thus its complement is deep exactly for $q \equiv 2 \pmod{3}$. The setwise stabilizer of the ordered normal form is the split dihedral group of order $2(q - 1)$, so the orbit size is

$$\frac{q(q^2 - 1)}{2(q - 1)} = \frac{q(q + 1)}{2}.$$

For a conjugate pair Frobenius interchanges the fixed points and inverts the multiplier; descent is instead $q \equiv 2 \pmod{3}$. The nonsplit dihedral stabilizer has order $2(q + 1)$ and the orbit size is $q(q - 1)/2$.

Now suppose the characteristic is three. The second Hasse derivative of

$$\nu(t) = (1, t, t^2, t^3, t^4)$$

is $(0, 0, 1, 3t, 6t^2) = e_2$, including in the reversed infinity chart. Thus every osculating plane contains $n = [e_2]$. Substitution shows that n is $\mathrm{PGL}_2(q)$ -fixed and $W_n = \langle T^3, U^3 \rangle$, so it is deep.

A remaining separable cyclic cover has one wild totally ramified point. Sending it to infinity and comparing coefficients under a nontrivial translation gives

$$W_f = \langle U^3, T^3 + \alpha T^2 U + \beta T U^2 \rangle, \quad \alpha = 0, \quad \kappa^2 = -\beta.$$

It is therefore represented by

$$f_a = e_2 - a e_4, \quad W_{f_a} = \langle U^3, T^3 + a T U^2 \rangle.$$

The affine members are $z^3 + az + s$. Such a member has three rational roots exactly when the additive map $z \mapsto z^3 + az$ has a nonzero rational kernel, equivalently when $-a$ is a square. Hence the nonsquare parameters form the deep orbit. Its stabilizer is $t \mapsto \pm t + \beta$, of order $2q$, and orbit-stabilizer gives $(q^2 - 1)/2$. $\square$

Lemma 4.6 (S_3 stratum). If ϕ_f has geometric monodromy S_3 and $q \geq 23$, then the pencil contains a completely split squarefree cubic.

Proof. By Proposition 4.4, components of Y_f correspond to monodromy orbits on distinct ordered roots. The S_3 action is two-transitive, so Y_f is geometrically integral and has arithmetic genus one. Aubry–Perret gives

$$\#Y_f(\mathbb{F}_q) \geq q - 2\sqrt{q}.$$

The diagonal costs

$$Y_f \cdot \Delta = (2, 2) \cdot (1, 1) = 4.$$

Riemann–Hurwitz gives different degree four for the separable cubic map, hence at most four branch values. A singular cubic has at most two distinct roots and at most two ordered off-diagonal pairs, so the branch deletion costs at most eight. If $q - 2\sqrt{q} > 12$, a rational off-diagonal unramified point remains. It gives two distinct rational roots of a squarefree cubic, whose third root is rational. The inequality holds for every prime power $q \geq 23$. $\square$

Proposition 4.7 (Exhaustion and finite bridge). The preceding cases exhaust every redundancy-five pencil for $q \geq 23$. Certificate R5 closes exactly

$$q \in \{7, 8, 9, 11, 13, 16, 17, 19\}.$$

It finds sporadic S_3 -stratum orbits exactly at $q = 7, 8, 9, 11, 13, 17, 19$ and none at $q = 16$.

Proof. The gcd degree is zero, one, or two. Propositions 4.2 and 4.3 settle the positive-gcd cases. In the trivial-gcd case the cubic map is inseparable or separable. The inseparable case occurs only in characteristic three and the overlap equations give the all-cube nucleus pencil. A separable transitive cubic has geometric monodromy C_3 or S_3 , settled by Lemmas 4.5 and 4.6. The stated finite residue, canonical representatives, stabilizers, histograms, and Frobenius links are exactly the domain of Certificate R5. $\square$

The orbit sizes in Table 4 are a compact inventory, not orbit identifiers. The electronic C491 certificate records for each row a canonical normalized representative, stabilizer, complete cubic-pencil member histogram, and Frobenius image; equality with one of those canonical PGL_2 orbits is the finite classification criterion. Thus repeated sizes denote distinct certified orbits.

Table 4: Complete sporadic orbit inventory at redundancy five. Entries are $\mathrm{PGL}_2(q)$ orbit sizes; exponents denote multiplicity.

q	number of sporadic points	orbit sizes
7	644	$28, 112, 168^3$
8	756	252^3
9	900	$180, 360^2$
11	990	$330, 660$
13	728	$182, 546$
17	1224	1224
19	570	570

5 Coherent polar induction

5.1 Spaces, markers, and bad schemes

Fix a field k , a two-dimensional k -space E , and

$$S_n = \mathbb{P}(\Gamma^n E).$$

For $\lambda \in E^\vee$, define divided-power contraction by

$$\langle \iota_\lambda f, g \rangle = \langle f, \lambda g \rangle.$$

Then

$$g \in W_{\iota_\lambda f} \iff \lambda g \in W_f. \quad (1)$$

The lower witness lifts squarefreely only when $\lambda \nmid g$. Thus the factor selected for contraction must remain as a marked forbidden root.

Definition 5.1 (Contraction and marker spaces). Let $C_f : E^\vee \rightarrow \Gamma^{n-1} E$ be $\lambda \mapsto \iota_\lambda f$, and put $U_f = \mathbb{P}(E^\vee) \setminus \mathbb{P}(\ker C_f)$. The first-polar rational map is

$$\Phi_f : U_f \longrightarrow S_{n-1}, \quad [\lambda] \longmapsto [\iota_\lambda f].$$

Its image is a point or a line. For $j \geq 1$, let

$$\mathrm{Conf}_j(\mathbb{P}(E^\vee)) = \{(\lambda_1, \dots, \lambda_j) : \lambda_i \neq \lambda_\ell \text{ for } i \neq \ell\}.$$

The ordered marker space $U_{f,j}$ is the open subset on which every successive contraction is nonzero. The universal j -step polar flag is the map

$$(\lambda_1, \dots, \lambda_j) \longmapsto ([\iota_{\lambda_1} f], \dots, [\iota_{\lambda_j} \cdots \iota_{\lambda_1} f]).$$

In an affine coordinate r the first map is

$$\Phi_f(r) = (a_1 - ra_0, \dots, a_n - ra_{n-1}), \quad \Phi_f(\infty) = (a_0, \dots, a_{n-1}).$$

Definition 5.2 (Lower splitting package). At level (n, j) a lower splitting package consists of:

- (i) a finite stratification $X_{n-j}^{\mathrm{split}} = \coprod_\alpha X_\alpha$ of the ordered squarefree splitting incidence over S_{n-j} ;

- (ii) for each α , a specified identity or Frobenius-twisted geometrically integral root cover $Y_\alpha \rightarrow X_\alpha$, with genus bound g_α and point bound

$$\#Y_\alpha(\mathbb{F}_q) \geq q + \kappa_\alpha - 2g_\alpha\sqrt{q};$$

- (iii) a closed bad scheme $B_{n-j} \subset S_{n-j}$ containing the loci on which the chosen cover is not available;
- (iv) branch, diagonal, old-marker, new-marker, and self-collision divisors whose total degree on Y_α is at most δ_α ;
- (v) bounds b and c for the pullbacks of B_{n-j} and the finite self-collision divisor to a non-contained first-polar line.

The index α names a geometric stratum together with its constant field and Frobenius twist. All degrees are geometric degrees after the displayed pullback. “Computed” means that equations for B_{n-j} and every divisor are obtained by finite contraction, elimination, factorization, and rank tests over the coefficient ring.

The construction is summarized in Figure 2. The upper branch is the level-specific component calculation; the lower branch is the general transverse argument.

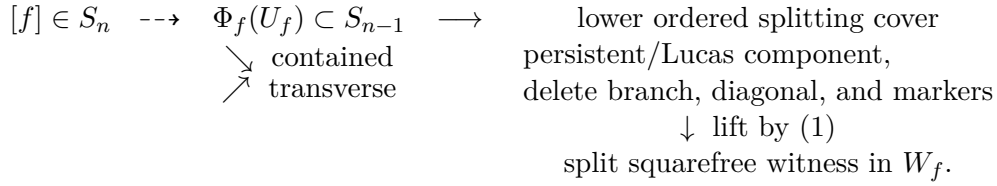


Figure 2: Coherent polar induction. The contained branch is a level-specific component theorem; only the transverse branch is controlled by the general point-count argument.

Theorem 5.3 (Polar-flag construction). The maps of Definition 5.1 commute with field extension and the natural $\mathrm{GL}(E)$ action. They include the infinity chart displayed above. If $g \in W_{\iota_{\lambda_j} \dots \iota_{\lambda_1} f}$, then $\lambda_1 \dots \lambda_j g \in W_f$; the lift is squarefree exactly when g is squarefree and avoids every marker λ_i .

Proof. Contraction is defined by the perfect divided-power/symmetric-power pairing, so it commutes with base change and change of basis. Repeated application of (1) gives the membership assertion. Since the λ_i are distinct on the configuration space, their product is squarefree; the product with g is squarefree precisely when g is squarefree and none of the λ_i divides it. The coordinate formula at infinity follows by evaluating C_f on the second basis covector. $\square$

Definition 5.4 (Contained-component assertion). For a specified lower package, $\mathrm{CC}(n, j)$ is the scheme-theoretic assertion that every $f \in S_n$ for which $\Phi_f^* B_{n-j}$ or a marker-collision section vanishes identically belongs to the explicitly listed closed subscheme $\mathcal{P}_n \cup \mathcal{M}_n$. An identically zero collision differential is part of $\mathcal{M}_n$; it is never counted as a finite divisor.

This is a level-specific theorem obligation, not an assumption hidden inside the point-count argument. Sections 6 and 7 state the contained-component calculations used at redundancies six through nine. No assertion $\mathrm{CC}(n, j)$ is made for an arbitrary degree.

Theorem 5.5 (Effective transverse induction). Fix a lower splitting package as in Definition 5.2 and suppose $\text{CC}(n, j)$ has been proved for that package. Put

$$\mathcal{H}_\kappa(g, \delta) = \left\lfloor \left(g + \sqrt{g^2 + \delta - \kappa} \right)^2 \right\rfloor + 1$$

and assume $\delta \geq \kappa - g^2$. If

$$q \geq \max_{\alpha} \mathcal{H}_{\kappa_{\alpha}}(g_{\alpha}, \delta_{\alpha}) \quad \text{and} \quad q + 1 > b + c,$$

then every split-free Hankel system belongs to the contained scheme $\mathcal{P}_n \cup \mathcal{M}_n$.

Proof. For a system outside $\mathcal{P}_n \cup \mathcal{M}_n$, the assertion $\text{CC}(n, j)$ makes the pulled-back bad and self-collision schemes finite of total degree at most $b + c$. Since $\#\mathbb{P}^1(\mathbb{F}_q) = q + 1$, choose a rational contraction parameter outside them. On the corresponding lower stratum the recorded point bound gives

$$q + \kappa_{\alpha} - 2g_{\alpha}\sqrt{q} > \delta_{\alpha},$$

so there is a rational point outside all deletions. It yields a split squarefree lower member avoiding every marker. Equation (1) lifts it to a split squarefree member upstairs, contradicting split-freeness. $\square$

Remark 5.6 (Point-count convention). The correction κ records the exact lower bound available for the chosen model. The singular cubic fiber square of Lemma 4.6 uses $\kappa = 0$. The normalized genus-one covers in the fixed-level applications use $\kappa = 1$. Thus the redundancy-eight triple $(g, \delta, \kappa) = (1, 30, 1)$ gives the real cutoff $(1 + \sqrt{30})^2 < 42$, whose first prime power is 43; the redundancy-nine triple $(1, 36, 1)$ gives the strict cutoff $q > 49$, whose first prime power is 53.

Table 5: Normalized point-count thresholds used in this paper.

Application	g	δ	κ	integer lower bound	first prime power
R5 cubic fiber square	1	12	0	$q \geq 22$	23
R6 marked cover	1	18	1	$q \geq 28$	29
R7 marked cover	1	24	1	$q \geq 35$	37
R8 three-marker cover	1	30	1	$q \geq 42$	43
R9 four-marker cover	1	36	1	$q \geq 50$	53

5.2 Contained-component calculations

The ordinary rank-two carrier admits a uniform contained-line calculation. This does not classify other lower bad schemes, but it closes the persistent part of every level-specific component gate.

Proposition 5.7 (Contained rank-two polar lines). Let $n \geq 5$, let $f \in \mathbb{P}(\Gamma^n E)$, and let

$$A = C_f^{(2)} : \text{Sym}^{n-2} E^{\vee} \longrightarrow \Gamma^2 E$$

be the three-row catalecticant. Assume the contraction map $E^{\vee} \rightarrow \Gamma^{n-1} E$ is injective, so the first-polar map is defined on all of $\mathbb{P}(E^{\vee})$. Then

$$\Phi_f(\mathbb{P}(E^{\vee})) \subset \{b : \text{rank } C_b^{(2)} \leq 2\} \iff \text{rank } A \leq 2$$

scheme-theoretically and in every characteristic. If $\text{rank } A = 3$, the rank-drop parameters form a subscheme of $\mathbb{P}(E^{\vee})$ of degree at most three.

Proof. For a nonzero $\lambda \in E^\vee$, multiplication identifies $\mathrm{Sym}^{n-3} E^\vee$ with the hyperplane

$$H_\lambda = \lambda \mathrm{Sym}^{n-3} E^\vee \subset \mathrm{Sym}^{n-2} E^\vee,$$

and functoriality gives

$$C_{\iota_\lambda f}^{(2)} = A|_{H_\lambda}.$$

The forward implication is immediate. Conversely, suppose $\mathrm{rank} A = 3$ and put $K = \ker A$, so $\dim K = n - 4$. If the lower rank is at most two, rank-nullity on the $(n - 2)$ -dimensional space H_λ gives

$$\dim(K \cap H_\lambda) \geq (n - 2) - 2 = n - 4 = \dim K.$$

Thus $K \subset H_\lambda$. Containment for every geometric $[\lambda]$ would imply

$$K \subset \bigcap_{[\lambda] \in \mathbb{P}(E_k^\vee)} \lambda \mathrm{Sym}^{n-3} E_k^\vee = 0,$$

because a nonzero binary form cannot be divisible by every linear form. This contradicts $\dim K = n - 4 > 0$.

The maximal minors of $C_{\iota_\lambda f}^{(2)}$ are homogeneous cubics in λ . Vanishing at every geometric point is therefore identical vanishing, proving the scheme-theoretic assertion. In the non-contained case at least one cubic is nonzero, and the common rank-drop scheme lies in its degree-three zero divisor. $\square$

After rank one and rational split secants are removed as shallow, the upper rank-two locus consists of the persistent tangent and conjugate-secant carriers. On a persistent fiber the nonsplit torus acts by $z \mapsto z^n$, giving T/T^n modulo inversion and Frobenius, while the tangent unipotent cocycle is $z \mapsto z + nu$.

Modular carriers are equally explicit. If $\mathcal{N}$ is a nucleus in the lower divided-power module, the complete space whose first-polar line lies in $\mathcal{N}$ is

$$\mathcal{M}_n(\mathcal{N}) = \mathbb{P} \ker \left(\Gamma^n E \longrightarrow \mathrm{Hom}(E^\vee, \Gamma^{n-1} E / \tilde{\mathcal{N}}) \right).$$

Lucas' theorem computes the nucleus coordinates, and this displayed kernel computes every coherent lift without an ambient syndrome census.

Remark 5.8 (Scope of the contained calculations). The modular contraction kernel is an exact finite linear-algebra construction. Proposition 5.7 supplies the ordinary persistent component uniformly, but other lower bad schemes and marker collisions still require level-specific calculations. Neither the proposition nor Theorem 5.5 classifies those additional components.

6 Redundancies six and seven: complete and gated results

6.1 Redundancy six

For

$$f = (a_0 : \cdots : a_5) \in \mathbb{P}(\Gamma^5 E), \quad W_f = \ker \begin{pmatrix} a_0 & a_1 & a_2 & a_3 & a_4 \\ a_1 & a_2 & a_3 & a_4 & a_5 \end{pmatrix},$$

the system W_f is a net of quartics off the quintic normal rational curve. It is split-free exactly when it has no completely split squarefree quartic.

Proposition 6.1 (Persistent net and orbit law). If $\gcd(W_f) = Q$ has degree two, then

$$W_f = Q \mathrm{Sym}^2 E^\vee.$$

The deep persistent locus has size $q(q+1)^2/2$. On the tangent fiber the unipotent coordinate is $x \mapsto x + 5u$; on the irreducible quadratic fiber the norm-one torus acts through $z \mapsto z^5$, with inversion and coefficient Frobenius acting on T/T^5 .

Proof. For fixed Q , the syndromes annihilating $Q \operatorname{Sym}^3 E^\vee$ form a projective line. If Q is irreducible, all $q + 1$ points remain; if $Q = \lambda^2$, one endpoint has dependent Hankel rows, leaving q points. Hence

$$q(q+1) + \frac{q(q-1)}{2}(q+1) = \frac{q(q+1)^2}{2}.$$

Normalize a square to U^2 . The nondegenerate fiber is $[e_4 + xe_5]$, and direct degree-five substitution gives $x \mapsto x + 5u$. For an irreducible Q , its stabilizer identifies the $q + 1$ -point fiber with $T = \{z \in \mathbb{F}_{q^2}^\times : z^{q+1} = 1\}$ and acts by fifth powers; the normalizer inverts z , while coefficient Frobenius multiplies T/T^5 by the characteristic. This proves the stated orbit law and the characteristic-five tangent splitting. $\square$

Proposition 6.2 (First-polar line and secant degree). For $r \in \mathbb{F}_q$ put

$$b(r) = (a_1 - ra_0, \dots, a_5 - ra_4), \quad b(\infty) = (a_0, \dots, a_4).$$

Then

$$(T - rU)g \in W_f \iff g \in W_{b(r)}.$$

If W_f has trivial gcd, its polar line meets the lower secant cubic in degree at most three.

Proof. The first equivalence follows by expanding the two Hankel equations of $(T - rU)g$. Let

$$C_f = \begin{pmatrix} a_0 & a_1 & a_2 & a_3 \\ a_1 & a_2 & a_3 & a_4 \\ a_2 & a_3 & a_4 & a_5 \end{pmatrix}.$$

The lower catalecticant equals

$$C_f \begin{pmatrix} -r & 0 & 0 \\ 1 & -r & 0 \\ 0 & 1 & -r \\ 0 & 0 & 1 \end{pmatrix}.$$

Cauchy–Binet gives the secant determinant

$$D_f(r) = -\Delta_{012}r^3 + \Delta_{013}r^2 - \Delta_{023}r + \Delta_{123}.$$

It vanishes identically exactly when $\operatorname{rank} C_f \leq 2$, equivalently when W_f has quadratic gcd. In the trivial-gcd case it therefore has at most three zeros. $\square$

Proposition 6.3 (Cyclic and collision degrees). For a trivial-gcd net:

- (i) in characteristic different from two, the lower tame cyclic carrier is a binomially rescaled Veronese surface of degree four and contains no polar line;
- (ii) the pointed ramification condition has degree at most six and does not vanish identically;
- (iii) an S_3 lower pencil with the marked root deleted has total deletion degree at most eighteen.

Proof. In syndrome coordinates the tame cyclic surface is

$$\{\operatorname{diag}(1, 4, 6, 4, 1)^{-1}[Q^2] : [Q] \in \mathbb{P}(\operatorname{Sym}^2 E^\vee)\}.$$

Outside characteristic two it is the Veronese surface, of degree four, and contains no line. For a basis F_0, F_1, F_2 of the $g_4^2 = W_f$, pointed collision is the rank-one condition

$$\text{rank} \begin{pmatrix} F_0(r) & F_1(r) & F_2(r) \\ F'_0(r) & F'_1(r) & F'_2(r) \end{pmatrix} \leq 1.$$

Its Wronskian divisor has degree six. In characteristic at least five it cannot vanish identically because the degrees are below the characteristic. In characteristics two and three, either all derivatives vanish or the derivative row is a rational multiple of the value row; the Hankel overlap excludes the first alternative, and a gcd argument forces a common factor in the second.

Finally the R5 off-diagonal curve has the original deletion degree twelve. The unique cubic through a specified marker contributes at most six ordered pairs, giving $\delta = 18$ and

$$q - 2\sqrt{q} > 18.$$

Its first prime-power solution is 29. □

Proposition 6.4 (Modular contained components). In characteristic three the lower wild carrier is the degree-four cone

$$\text{Join}(e_2, C_4),$$

and no polar line of a trivial-gcd net is contained in it. In characteristic two the cyclic carrier is the plane

$$\Pi_{\text{cyc}} = \mathbb{P}\langle e_1, e_2, e_3 \rangle,$$

whose unique contained polar-line component is

$$\mathcal{N}_3 = \mathbb{P}\langle e_2, e_3 \rangle.$$

Proof. Every line on the wild cone is a ruling. If two consecutive Hankel rows lie in a ruling $\langle e_2, \nu(t) \rangle$, the overlap equations give $f = \mu\nu(t)$, including $f = \mu e_5$ at infinity. Thus no trivial-gcd polar line is contained.

In characteristic two, direct substitution gives

$$g \cdot e_2 = [0 : \delta\gamma : \alpha\delta + \beta\gamma : \alpha\beta : 0],$$

whose closure is Π_{cyc} . The consecutive-row overlap places a polar line in this plane exactly when $a_0 = a_1 = a_4 = a_5 = 0$, namely when $f \in \mathbb{P}\langle e_2, e_3 \rangle$. □

Proposition 6.5 (Binary nucleus arithmetic). The scheme $\mathcal{N}_3(\mathbb{F}_q)$ is one projective-semilinear orbit of $q + 1$ points. It is split-free over $q = 2^m$ exactly when m is odd.

Proof. For the representative e_3 ,

$$W_{e_3} = \langle 1, t, t^4 \rangle.$$

The root differences of $A + Bt + Ct^4$ lie in the kernel of $u \mapsto u^4 + (B/C)u$. Four distinct rational roots exist exactly when all roots of $u^3 = B/C$ are rational, equivalently when $3 \mid q - 1$, which for $q = 2^m$ means m is even. The stabilizer of e_3 is the Borel of order $q(q - 1)$, so the orbit has $q + 1$ points; its lower-unipotent transforms and endpoint are precisely $\mathcal{N}_3(\mathbb{F}_q)$. □

The covering-radius gate is complete: Seroussi–Roth gives $\rho(\text{PRS}(q - 5)) = 5$ for $q \geq 9$, while Certificate R6 checks it directly at $q = 7, 8$.

Theorem 6.6 (Redundancy six). For every prime power $q \geq 7$, the deep syndromes of $\text{PRS}(q - 5)$ are the persistent stratum of size $q(q + 1)^2/2$, together with:

- (i) respectively 18, 11, 4, 2, 1 exceptional $\mathrm{PGL}_2(q)$ orbits at $q = 7, 8, 9, 11, 13$;
- (ii) one characteristic-two nucleus orbit of size $q + 1$ when $q = 2^m$ and odd $m \geq 5$.

There are no other exceptional orbits. The complete persistent orbit law is T/T^5 modulo inversion and Frobenius, with a tangent fixed/nonzero split exactly in characteristic five.

Proof. Propositions 6.1–6.5 classify the common-factor and contained components and exclude a trivial-gcd exception in the conceptual ranges: characteristic at least five from $q \geq 29$, characteristic three from its next unchecked field $q = 81$, and characteristic two outside $\mathcal{N}_3$ from $q \geq 32$. Certificate R6 exhausts

$$q \in \{7, 8, 9, 11, 13, 16, 17, 19, 23, 25, 27\},$$

finds exceptional projective-orbit counts 18, 11, 4, 2, 1 only at the five displayed fields, and no further orbit at the remaining fields. Together with the separate radius gate, this covers every prime power $q \geq 7$. $\square$

The five small tables are exhaustive computations. Their semilinear orbit counts are 18, 5, 2, 2, 1; a collision-energy invariant plus, in odd characteristic, the quintic root-divisor type separates the classes up to precisely Frobenius fusion. The C498 census certificate supplies a canonical representative, stabilizer, member histogram, and Frobenius link for every projective orbit, while the normal-form certificate supplies the collision-energy and root-divisor fingerprint. Orbit counts alone are not used as identifiers.

6.2 Redundancy seven

For

$$f = (a_0 : \dots : a_6) \in \mathbb{P}(\Gamma^6 E), \quad W_f = \ker \begin{pmatrix} a_0 & a_1 & a_2 & a_3 & a_4 & a_5 \\ a_1 & a_2 & a_3 & a_4 & a_5 & a_6 \end{pmatrix},$$

the system W_f is a four-dimensional web of quintics. For an affine marker r the contractions are

$$b(r) = (a_1 - ra_0, \dots, a_6 - ra_5).$$

Coefficient collection gives

$$(t - r)g \in W_f \iff g \in W_{b(r)};$$

the lift is squarefree exactly when $g(r) \neq 0$.

Proposition 6.7 (Two-marker lower cover). Let x be a forbidden root on a trivial-gcd quartic net. On the generic S_3 bottom fiber, the diagonal and branch divisors and the two marked-root fibers have total degree at most

$$12 + 6 + 6 = 24.$$

Hence $q - 2\sqrt{q} > 24$, and in particular every prime power $q \geq 37$, produces a split lower quartic avoiding x .

Proof. The R5 off-diagonal curve is geometrically integral of arithmetic genus one and has base deletion degree twelve. Each marker belongs to a unique cubic fiber, which contains at most six ordered pairs. Delete those two fibers and apply the $\kappa = 0$ bound. $\square$

Proposition 6.8 (Exact-gcd-one avoidance). Suppose the lower net is $\ell_x U$, where $U = \ker \lambda \subset \mathrm{Sym}^3 E^\vee$ has dimension three and $x \neq r$. For $q \geq 16$ it contains a split cubic avoiding both x and r .

Proof. Put $S = \mathbb{P}^1(\mathbb{F}_q) \setminus \{x, r\}$. For ordered distinct $u, v \in S$, the map $w \mapsto \lambda(\ell_u \ell_v \ell_w)$ is linear. If it vanishes identically, choose any remaining w ; otherwise it has one rational zero. The equations $w = x, r, u, v$ have respective bidegrees $(1, 1), (1, 1), (2, 1), (1, 2)$ in (u, v) . None is identically zero: the first two would add a common factor, and the last two are excluded because double-root cubics span $\text{Sym}^3 E^\vee$. A nonzero bidegree- (a, b) divisor on $\mathbb{P}^1 \times \mathbb{P}^1$ has at most $(a + b)(q + 1)$ rational points. Thus the bad pairs number at most $10(q + 1)$, whereas $(q - 1)(q - 2) > 10(q + 1)$ for $q \geq 16$. $\square$

Proposition 6.9 (Marked self-collision). After fixed factors are removed, the self-collision divisor of the moving g_5^3 has degree at most eight.

Proof. It is the ramification divisor of a separable g_5^3 , whose degree is $(3 + 1)(5 - 3) = 8$. If the series were inseparable, it would factor through Frobenius and lie in the p -th-power subspace of quintics; that subspace has dimension at most three, smaller than the four-dimensional Hankel series. $\square$

The characteristic-two nucleus line at redundancy six has exactly one coherent lift, the central point e_3 , with

$$W_{e_3} = \langle 1, t, t^4, t^5 \rangle.$$

It is deep for odd m ; for even m , the homogenization of $t^4 + t$ has the five roots $\mathbb{P}^1(\mathbb{F}_4)$.

Proposition 6.10 (Central binary lift). The complete inverse image of the R6 nucleus line under consecutive sextic contraction is $[e_3]$. It is split-free over $\mathbb{F}_{2^m}$ exactly when m is odd.

Proof. Putting both consecutive rows in $\mathbb{P}\langle e_2, e_3 \rangle$ gives $a_0 = a_1 = a_2 = a_4 = a_5 = a_6 = 0$. Thus only e_3 remains, with $W_{e_3} = \langle 1, t, t^4, t^5 \rangle$. For odd m , contraction at a root of a hypothetical split quintic would give a split quartic on the R6 nucleus line, contradicting Proposition 6.5. For even m , $\mathbb{F}_4 \subset \mathbb{F}_q$ and the homogenization of $t^4 + t$ has its four affine $\mathbb{F}_4$ roots and the simple root at infinity. $\square$

Corollary 6.11 (Degree-six contained component). The assertion $\text{CC}(6, 1)$ for the lower rank-two carrier holds in every characteristic. Its contained sextics have upper catalecticant rank at most two; after the shallow rank-one and rational secant loci are removed, they are exactly the persistent tangent and conjugate-secant families.

Proof. Apply Proposition 5.7 with $n = 6$. The three rational types of the quadratic recurrence are split squarefree, repeated rational, and irreducible. They give respectively the shallow rational secant, tangent, and conjugate-secant loci. The remaining lower binary nucleus line has the separate complete lift computed in Proposition 6.10. $\square$

Theorem 6.12 (Redundancy-seven classification). Certificate R7 unconditionally classifies every prime power $q \leq 32$ in its displayed domain. For every prime power $q \geq 13$, the split-free syndromes at redundancy seven are the persistent stratum, together with the central nucleus point when $q = 2^m$ and m is odd. At $q = 7, 8, 9, 11$ the additional split-free orbit-size profiles are:

7	56, 84^5 , 112^2 , 168^{45} , 336^{141}
8	63, 72, 84^3 , 168^4 , 252^{24} , 504^{86}
9	180^3 , 240^6 , 360^{18} , 720^{27}
11	264^2 , 440, 660^2 .

This is the complete all-field split-free classification. Since Seroussi–Roth gives $\rho(\text{PRS}(q - 6)) = 6$ for every prime power $q \geq 11$, it is also the complete deep-hole classification in that range. At $q = 7, 8, 9$ the table is not promoted to a code deep-hole classification without a separate covering-radius calculation. The persistent orbit law is T/T^6 modulo inversion and Frobenius, with tangent splitting in characteristics two and three.

Proof. For $q \geq 37$, Propositions 6.7–6.9 give the transverse witness after at most three catalecticant intersections, eight self-collisions, and, in characteristic two, one intersection with the nucleus line. Corollary 6.11 leaves exactly the persistent locus and Proposition 6.10. Certificate R7 covers

$$q \in \{7, 8, 9, 11, 13, 16, 17, 19, 23, 25, 27, 29, 31, 32\}$$

by an independent five-secant replay. The radius theorem promotes the split-free list only from $q \geq 11$. $\square$

For the four finite rows the C509 certificate contains the canonical representative, stabilizer, persistent/central flags, and Frobenius link of every orbit. The displayed size profile is only a compact summary; repeated sizes are distinguished by those canonical records.

The $q = 19$ lower net $\langle 1, t^3, t^4 \rangle$ has six split members, all containing the marked point at infinity, yet no coherent sextic polar line lies in its orbit. This is the concrete reason Theorem 5.5 is formulated for parameterized flags rather than a set of bad lower fibers.

Table 6: Field-range closure ledger. “Certificate” means a complete finite classification with canonical representatives, not merely an orbit count.

Result	prime powers	closure mechanism
R5	7, 8, 9, 11, 13, 16, 17, 19 $q \geq 23$	direct census and independent replay cubic-cover geometry, Lemma 4.6
R6	7, 8, 9, 11, 13, 16 17, 19, 23, 25, 27 $q \geq 29$	direct syndrome/radius census finite structural bridge certificate marked-cover point bound and contained components
R7	7, 8, 9, 11 13, 16, 17, 19, 23, 25, 27, 29, 31, 32 $q \geq 37$	direct split-free census coherent-polar finite bridge certificate marked-cover point bound and contained components

The R7 rows classify split-free syndromes. Conversion to code deep-hole directions uses the separate covering-radius gate and is made only for $q \geq 11$. The public certificate schema and exact replay boundaries are given in the supplement; the table does not promote orbit-size profiles to identifiers.

7 Fixed-level high-field theorems

7.1 Redundancy eight

Proposition 7.1 (Generic three-marker bottom cover). After three contractions, the generic bottom object is the R5 off-diagonal $(2, 2)$ curve. The diagonal and branch deletion costs twelve and three marked roots cost eighteen, so $\delta = 30$. The inequality

$$q + 1 - 2\sqrt{q} > 30$$

holds at every prime power $q \geq 43$. The first-polar line has transverse/collision budget

$$3 + 1 + 10 = 14,$$

where ten is the ramification degree of a moving g_6^4 .

Proof. Markers do not change the bottom curve or its geometric monodromy; each removes at most the six ordered pairs in its unique cubic fiber. The point bound is the $\kappa = 1$ row of Table 5. A noncontained line meets the lower rank-two catalecticant carrier in degree at most three, and the lower binary central point in at most one parameter. After fixed factors

are removed, self-collision is ramification of a separable g_6^4 , of degree $(4 + 1)(6 - 4) = 10$. Separability follows because the p -th-power subspace has dimension smaller than the five-dimensional series. $\square$

Definition 7.2 (Three-marker lower-package gate). The assertion $\text{LP}(6, 1)$ says that the pointed R7 splitting incidence, after the persistent rank-two carrier, the binary central point, and the marked-collision divisor are removed, has the geometrically integral identity twists and deletion bounds used in Proposition 7.1. It includes equations and monodromy checks for the recursive cyclic, wild, inseparable, gcd, and branch strata; a finite-field regression is not a substitute for those geometric data.

Proposition 7.3 (Pointed lower package). The assertion $\text{LP}(6, 1)$ holds. On the outer first-marker line its exceptional divisor has degree at most fifteen; on the recursive second-marker line the corresponding divisor has degree at most nineteen. Its only positive-genus identity twist is the geometrically integral off-diagonal $(2, 2)$ curve of genus at most one with deletion degree 30.

Proof. We print the recursive strata. For a pointed sextic $f = (a_0, \dots, a_6)$ with old marker x , contract successively:

$$b_i(r) = a_{i+1} - ra_i, \quad c_i(r, s) = b_{i+1}(r) - sb_i(r).$$

For $c = (c_0, \dots, c_4)$ put

$$H_c = \begin{pmatrix} c_0 & c_1 & c_2 & c_3 \\ c_1 & c_2 & c_3 & c_4 \end{pmatrix}.$$

On the rank-two-kernel chart, if p_0, p_1 span $\ker H_c$, the bottom ordered-root curve is

$$G_c(u, v) = \frac{p_0(u)p_1(v) - p_1(u)p_0(v)}{[u, v]} = 0. \quad (25)$$

The rank-at-most-two secant carrier, including the gcd-two and rank-drop boundary strata, is

$$D(c) = \det \begin{pmatrix} c_0 & c_1 & c_2 \\ c_1 & c_2 & c_3 \\ c_2 & c_3 & c_4 \end{pmatrix} = 0. \quad (26)$$

For an exact common root z , the gcd-one incidence is cut out by

$$\text{rank} \begin{pmatrix} H_c & \\ 1 & z & z^2 & z^3 \end{pmatrix} \leq 2, \quad (27)$$

with the evident homogeneous row at infinity. After substitution, (26) has degree at most three in s , while each 3-minor in (27) has degree at most two for fixed z .

The remaining reducible-monodromy carriers are explicit. Put $z_i = \binom{4}{i} c_i$. Outside characteristics two and three the cyclic closure is

$$[u : v : w] \longmapsto [u^2 : 2uv : v^2 + 2uw : 2vw : w^2], \quad (28)$$

a degree-four projected Veronese surface containing no line. In characteristic three it is the degree-four cone $\bowtie (e_2, C_4)$; its only lines are rulings, and the consecutive-row overlap excludes every ruling as a polar line. In characteristic two it is

$$\Pi_{\text{cyc}} = \mathbb{P}\langle e_1, e_2, e_3 \rangle, \quad c_0 = c_4 = 0, \quad (29)$$

whose unique contained polar-line component is $\mathcal{N}_3 = \mathbb{P}\langle e_2, e_3 \rangle$. The inseparable degeneration is the derivative-minor locus of p_0, p_1 and introduces no further stratum. Outside characteristics two and three a nonconstant degree-three map cannot factor through Frobenius. In characteristic three an inseparable cubic pencil is projectively $\langle X^3, Y^3 \rangle$; the two Hankel rows then force $c_0 = c_1 = c_3 = c_4 = 0$, the vertex e_2 of the wild cone. In characteristic two, after cancelling a nonzero pencil gcd, an inseparable ratio has degree at most two and is a rational function of t^2 . Homogenizing its numerator and denominator to cubics supplies the same linear factor to both, so it belongs to the exact-gcd-one boundary. Finally, the branch equation on this chart is $p'_0 p_1 - p_0 p'_1 = 0$. These carrier and branch loci are the scheme-theoretic images of the displayed incidences (equivalently their Fitting-minor ideals); the reversed homogeneous chart supplies infinity.

Off (26)–(29), the separable cubic map is noncyclic, so its geometric monodromy is S_3 . Transitivity on ordered pairs of distinct roots makes (25) geometrically integral. It has arithmetic genus one. The diagonal costs four, the degree-four different costs at most eight off-diagonal ordered pairs, and each of the three marker fibers x, r, s costs at most six. Thus

$$\delta \leq 4 + 8 + 3 \cdot 6 = 30. \quad (30)$$

Hasse–Weil supplies a surviving rational point for every $q \geq 43$.

It remains to show that the strata removed above do not hide another component. On an exact-gcd-one bottom stratum write the cubic pencil as $\ell_z Q$, with Q a basepoint-free pencil of quadratics. In the separable case its deck involution ι is rational, so its graph has $q + 1$ rational points. At most two are fixed and at most two ordered graph points involve each of z, x, r, s . A point remains for $q \geq 43$, giving three distinct roots outside the markers. The inseparable degree-two case is on the rank/boundary carrier.

This gives the required two-old-marker quartic lemma. Indeed, on the internal s -line the bad degrees are: three from (26), four from the cyclic/wild carrier (one in characteristic two), six from ramification of the moving g_4^2 , two for each fixed marker from (27), and two marker diagonals. Their total is at most

$$3 + 4 + 6 + 2 + 2 + 2 = 19 < q + 1. \quad (31)$$

The rank pullback cannot be the whole line: Proposition 6.2 identifies that containment exactly with quadratic gcd of the original net. The only possible contained cyclic line is the declared binary nucleus. For a fixed marker, not all minors in (27) can vanish identically without giving the original quartic net that fixed factor: otherwise the hyperplanes $W \cap \ker(\text{ev}_s)$ and $W \cap \ker(\text{ev}_x)$ agree for dense s , making the evaluation map constant. Choose one nonzero quadratic minor. For self-collision, take a generic pencil projection of the moving g_d^2 , $d \leq 4$. It is separable: in characteristic two the only dimension-three equality case is the pure-square space $\langle 1, t^2, t^4 \rangle$, which the consecutive Hankel overlap equations do not realize; all other p -power spaces have smaller dimension. Riemann–Hurwitz gives degree at most $2d - 2 \leq 6$, and every self-collision maps into this ramification divisor.

For the pointed sextic itself, the lower rank-two pullback costs three parameters and the binary nucleus costs one. If a contracted net has exact gcd ℓ_z , the condition $z = r$ maps into the ramification of a generic separable pencil projection of the moving g_d^3 , $d \leq 5$. Riemann–Hurwitz bounds it by $2d - 2 \leq 8$. The condition $z = x$ is again (27), now with the quartic evaluation row, and costs at most two because not all of its quadratic minors vanish off the fixed-marker boundary, by the same evaluation-hyperplane argument. Together with $r = x$, the internal divisor has degree

$$3 + 1 + 8 + 2 + 1 = 15. \quad (32)$$

If the contracted net has trivial gcd, the two-old-marker lemma applies. If it has exact gcd with $z \notin \{x, r\}$, the ordered-pair argument has three $(1, 1)$ and two $(2, 1)/(1, 2)$ bad equations,

at most $12(q+1) < (q-2)(q-3)$. The cases $z = x, r$ are precisely the excluded quadratic and ramification divisors. Lifting first by ℓ_s and then by ℓ_r gives a split squarefree quintic avoiding x .

The gcd has degree zero, one, or at least two, and a separable gcd-zero cubic cover has monodromy S_3 or C_3 ; hence the displayed list is exhaustive. No certificate or finite-field regression is used for this integrality assertion. Notice also that the outer and recursive parameter budgets 15 and 19 are already strictly below $q+1$ for $q \geq 19$; the threshold 43 is imposed solely by the genus-one deletion inequality (30), not by a hidden parameter-choice bound. $\square$

Proposition 7.4 (Modular lifts at degree seven). The consecutive-row Lucas calculation leaves no nonzero lift in characteristic two, the line $\mathbb{P}\langle e_2, e_5 \rangle$ in characteristic three, and the line $\mathbb{P}\langle e_3, e_4 \rangle$ in characteristic five. The latter two lines are shallow in the range $q \geq 43$.

Proof. Lucas' criterion applied to the consecutive binomial rows gives the three stated supports; the exact row reductions are replayed by Certificate R8. In characteristic five, the homogenization of $t^5 - t$ supplies six distinct projective roots and satisfies the two required middle-coefficient equations throughout the lift.

In characteristic three, $\mathrm{PGL}_2(q)$ is transitive on the rational points of $\mathbb{P}\langle e_2, e_5 \rangle$, so it suffices to treat e_2 . Choose nonzero a, b, c with no two equal up to sign and $a^2 + b^2 + c^2 = 0$. The nonsingular conic has $q+1$ points; deleting the three coordinate and six equality/antiequality sections costs at most eighteen, so a choice exists for $q \geq 27$. The reciprocal pairs $\pm a^{-1}, \pm b^{-1}, \pm c^{-1}$ give six distinct roots and the two required elementary-symmetric coefficients vanish. $\square$

Proposition 7.5 (Degree-seven contained components). The assertion $\mathrm{CC}(7, 1)$ for the structural components of Definition 7.2 holds in every characteristic. The rank-two component lifts to the upper rank-two carrier; the lower binary central point has no nonzero coherent lift; and the marked-collision scheme is never the whole polar line. The only additional nucleus lifts are

$$\mathbb{P}\langle e_2, e_5 \rangle \quad \text{in characteristic three,} \quad \mathbb{P}\langle e_3, e_4 \rangle \quad \text{in characteristic five,}$$

and both are shallow in the range $q \geq 43$.

Proof. The rank-two assertion is Proposition 5.7 with $n = 7$. For the binary central point e_3 , requiring both consecutive septic rows to be supported at its single coordinate permits a_3 in the first row and a_4 in the second; their overlap is zero. Thus its projective coherent lift is empty. The full nucleus-support overlap is exactly Proposition 7.4, which also supplies the shallow witnesses in characteristics three and five.

It remains to exclude an identically colliding series. Remove the fixed divisor from the five-dimensional W_f and let the moving degree be $d \leq 6$. If its projective map were inseparable in characteristic p , it would factor through Frobenius, so its section space would have dimension at most

$$\lfloor d/p \rfloor + 1 \leq 4,$$

contrary to $\dim W_f = 5$. A generic pencil projection is therefore a separable map of degree at most six. Every marked self-collision is a ramification point of that projection, so Riemann–Hurwitz bounds the collision divisor by $2d - 2 \leq 10$. It is finite and cannot be an additional contained component. $\square$

Remark 7.6 (Exact remaining R8 gate). Proposition 7.5 closes the contained-component problem, while Proposition 7.3 closes the pointed lower package. The certificate checks the nucleus overlaps, witnesses, and numerical budgets; the geometric integrality and recursive carrier equations are proved separately above.

Theorem 7.7 (Redundancy eight). For every prime power $q \geq 43$, the deep syndromes of $\text{PRS}(q-7)$ are exactly the persistent tangent and conjugate-secant families, of total size $q(q+1)^2/2$. Their orbit law is T/T^7 modulo inversion and Frobenius. The tangent family splits exactly in characteristic seven.

Proof. Outside the contained locus, Proposition 7.1 chooses a good first contraction, and Proposition 7.3 supplies a three-marker witness. Propositions 7.4 and 7.5 eliminate every nonpersistent contained lift. The imported covering-radius theorem promotes the resulting split-free classification to deep holes. $\square$

This theorem makes no assertion about additional deep orbits for $q < 43$.

7.2 Redundancy nine

Proposition 7.8 (Four-marker budgets). Four markers give deletion degree

$$12 + 4 \cdot 6 = 36,$$

and the first prime-power solution of $q + 1 - 2\sqrt{q} > 36$ is 53. The transverse/collision budget is at most

$$3 + 2 + 12 = 17.$$

Proof. The bottom curve contributes twelve diagonal/branch points and each marker its six ordered pairs. The first-polar line meets the persistent catalecticant carrier in degree at most three and the linear nucleus support in at most two parameters. Self-collision is ramification of a separable g_7^5 , of degree $(5+1)(7-5) = 12$. The normalized point bound is the $(g, \delta, \kappa) = (1, 36, 1)$ row of Table 5. $\square$

The only substantial modular carrier occurs in characteristic seven and is a projective binary-quartic space. Fixing a split quintic $P(t) = \sum_{i=0}^5 p_i t^i$ and a carrier point $h = (a_0, \dots, a_4)$, put

$$H_j = \sum_{i=0}^4 a_i p_{i+j}$$

with $p_k = 0$ outside $0 \leq k \leq 5$, and

$$D = H_0 H_2 - H_1^2, \quad N_s = H_{-1} H_2 - H_0 H_1, \quad N_u = H_{-1} H_1 - H_0^2.$$

Off $D = 0$, the unique residual quadratic is

$$Dt^2 - N_s t + N_u,$$

with branch polynomial $K = N_s^2 - 4N_u D$. The exact residual identities are proved next; the geometric exhaustion of transported slices is kept as a separate printed gate.

Proposition 7.9 (Residual quadratic identities). With $p_k = 0$ outside $0 \leq k \leq 5$, the two consecutive Hankel equations for

$$P(t)(t^2 - st + u)$$

are

$$H_0 - sH_1 + uH_2 = 0, \quad H_{-1} - sH_0 + uH_1 = 0.$$

On $D \neq 0$, Cramer's rule gives

$$s = N_s/D, \quad u = N_u/D.$$

The branch divisor is $K = 0$, fixed/residual collision is $\text{Res}(P, Dt^2 - N_s t + N_u) = 0$, and fixed-root collision is $\text{Disc}(P) = 0$.

Proof. Expand the two consecutive coefficients annihilated by the carrier quartic. The displayed linear system has determinant D , so Cramer's rule gives the residual factor. Its discriminant is K . The two resultants have their usual intrinsic collision meanings; consequently the construction commutes with scalar, PGL_2 , and Frobenius transport. $\square$

Proposition 7.10 (One-moving-root slice). Let $P = R(t)(t - x)$ with four fixed distinct roots. Then D, N_s, N_u have degree at most two in x , and K has degree at most four. If the squarefree reduction of K is nonconstant and not a square, the normalization of $y^2 = K(x)$ is geometrically integral of genus at most one.

Proof. Each H_j is affine in x , so its 2×2 minors are quadratic and K is quartic. In odd characteristic the squarefree nonsquare equation defines a geometrically integral double cover; the hyperelliptic genus formula gives $\lfloor (\deg K_{\mathrm{red}} - 1)/2 \rfloor \leq 1$. $\square$

Remark 7.11 (Six-slice and base-selection gates). Certificate R9 records six reduced discriminant polynomials for the squarefree quartic normal form, their unit gcd, and four multiple-root controls. The development report also states a nonzero rational base-selection polynomial of individual degree at most 24 and total degree at most 96. A refereeable proof must print the transporter, the six polynomials and a Bézout identity, the four reduced branch polynomials, and the nonzero base polynomial with its degree accounting. Until those data appear in the paper or immutable supplement, the six-slice component theorem and the $q \geq 343$ base selection are explicit gates rather than consequences of the certificate.

Proposition 7.12 (Good-slice deletion). Once a rational base with an integral slice is fixed, the moving/fixed diagonal, determinant, branch, four fixed-root collisions, and moving-root collision have degrees

$$4, 2, 4, 8, 4,$$

respectively. Their total is 22, so $q + 1 - 2\sqrt{q} > 22$ supplies a split witness.

Proof. The first three degrees follow from Proposition 7.10; each fixed-root collision is linear in the moving parameter on each of four residual sheets, and the final resultant has degree four. Removing their union from the normalized genus-at-most-one slice proves the claim. $\square$

Proposition 7.13 (Characteristic-seven finite bridges). At $q = 7$, the formal carrier has exactly 819 projective rootless quartics. At $q = 49$, Certificate R9-49 exhausts all

$$\frac{49^5 - 1}{48} = 5,884,901$$

projective carrier points and finds a split witness for each.

Proof. At $q = 7$, a split squarefree degree-seven projective divisor is the complement of one of the eight rational points. Inclusion–exclusion gives

$$\frac{7^5 - 1}{6} - 8\frac{7^4 - 1}{6} + 28\frac{7^3 - 1}{6} - 56\frac{7^2 - 1}{6} + 70\frac{7 - 1}{6} = 819.$$

The $q = 49$ clause is deliberately certificate-only: one lexicographic five-root search stops after every carrier point has a witness. The residual algebra is independently replayed, but there is no second exhaustive carrier implementation. $\square$

Proposition 7.14 (R9 persistent orbit law). Put $d = \gcd(8, q+1)$. The sigma-fiber projective orbits are the inversion orbits in $C_d = T/T^8$, and coefficient Frobenius acts by multiplication by the characteristic. The tangent fiber is transitive unless the characteristic is two, when its zero and nonzero parts are separate.

Proof. The nonsplit torus acts by eighth powers and its normalizer inverts the torus coordinate. Thus inversion-fixed and paired classes have stabilizers $2d$ and d , respectively; coefficient Frobenius is the p -power map. For $d = 8$, it fuses the two odd pairs exactly when $p \equiv \pm 3 \pmod{8}$. On the tangent fiber the unipotent action is $z \mapsto z + 8u$, which vanishes exactly in characteristic two. $\square$

Proposition 7.15 (Other modular lifts). The consecutive Lucas supports leave a line in characteristic five and the quartic carrier above in characteristic seven; no other characteristic has a nonzero coherent lift. The former line is shallow for every $q > 5$.

Proof. Lucas' theorem and the consecutive-row overlap give the two supports. In characteristic five, homogenize $t^5 - t$ to include its simple root at infinity and multiply by a linear factor $t - a$ with $a \notin \mathbb{F}_5$. The seven roots are distinct and the two required Hankel coefficients vanish. Such an a exists for every $q > 5$. $\square$

Theorem 7.16 (Conditional redundancy nine). Assume the six-slice and rational-base assertions of Remark 7.11, together with the degree-eight contained-component assertion. For every prime power $q \geq 53$, the deep syndromes of $\text{PRS}(q - 8)$ are exactly the persistent tangent and conjugate-secant families, of total size $q(q + 1)^2/2$. Their orbit law is T/T^8 modulo inversion and Frobenius; the tangent family splits exactly in characteristic two.

Proof. Proposition 7.8 handles every transverse nonmodular point. The assumed contained assertion leaves the persistent and modular supports. Proposition 7.15 eliminates characteristic five. For characteristic seven, the only powers below 343 are 7 and 49, both below the headline range; at every $q = 7^m \geq 343$, the assumed base-selection assertion and Proposition 7.12 produce a witness. The covering-radius theorem completes the promotion to deep holes. $\square$

Proposition 7.13 is a carrier boundary, not part of the headline classification: the $q = 7$ count is not a deep-hole statement for the inadmissible parameter $\text{PRS}(-1)$, and the $q = 49$ pass is not an ambient $\mathbb{P}^8$ census.

8 Characteristic two: the ordered Hessian

In characteristic two the divided binary-cubic discriminant becomes the doubled quadric $(AD + BC)^2$, so its pullback cannot distinguish generic splitting. The correct replacement retains an ordered residual root.

Let $\ell = \mathbb{P}\langle U, V \rangle \subset \mathbb{P}(\Gamma^3 E)$ and write

$$\text{Hess}^{\text{div}}(uU + vV) = N_u(u, v)X^2 + N_s(u, v)XY + D(u, v)Y^2.$$

The ordered-Hessian incidence

$$N_u(u, v)X^2 + N_s(u, v)XY + D(u, v)Y^2 = 0 \tag{2}$$

has bidegree $(2, 2)$ and arithmetic genus one. Off $N_s = 0$, its Artin–Schreier class is DN_u/N_s^2 .

The parameter space of cubic lines is $G(1, \Gamma^3 E)$. Let

$$\Sigma = \{\mathbb{P}(QE) : [Q] \in \mathbb{P}(\text{Sym}^2 E)\} \subset G(1, \Gamma^3 E)$$

be the Veronese surface of common-quadratic lines. In the ambient cubic space, let $\mathcal{Q}_0 = \{AD + BC = 0\} \subset \mathbb{P}(\Gamma^3 E)$. Its two families of lines give two Fano ruling conics in the

Grassmannian; one is the tangent-conic boundary of Σ , and we call the other $\mathcal{N}$. A root-compatible pullback is the family of lines obtained by fixing a split factor R and varying the final two linear factors in the consecutive Hankel contraction of a syndrome.

$$\begin{array}{ccc}
G(1, \Gamma^3 E) \supset \text{degenerate ordered-Hessian locus} & = & \Sigma \cup \text{Fano}_1(\mathcal{Q}_0) \\
\downarrow \text{root-compatible pullback} & & \\
\text{persistent component} & & \\
\cup & & \mathcal{N} : \text{no rank-two pullback.} \\
\text{Lucas-nucleus component} & &
\end{array}$$

Figure 3: Ordered-Hessian geometry. The degeneracy locus is computed in the Grassmannian first; root compatibility then selects the contained syndrome carriers and eliminates the complementary ruling.

Theorem 8.1 (Ordered-Hessian geometric degeneracy). After removing the forced vertical factors from intersections with the twisted cubic, the complete geometric degeneracy locus of (2) consists of:

- (i) the Veronese surface Σ of lines $\mathbb{P}(QE)$, on which the incidence is separably reducible; and
- (ii) the two Fano ruling conics of $\mathcal{Q}_0$, on which the root projection of (2) is inseparable.

The complementary ruling $\mathcal{N}$ has no nontrivial rank-two root-compatible Hankel pullback.

Proof. Once vertical and horizontal factors are removed, any separable factorization of a $(2, 2)$ curve must be $(1, 1) + (1, 1)$. The two factors are graphs of projectivities. Semisimple and unipotent normal forms show that only the lines QE occur. The ambient equation $AD + BC = 0$ gives two quadric rulings. Substituting the consecutive Hankel contraction forms into the complementary ruling forces overlapping triples of linear forms to be proportional, so the root-compatible line has rank at most one. $\square$

Remark 8.2 (Ordered-Hessian pullback gates). Two further statements are required for the advertised PRS containment corollary. First, the pullback of Σ must be proved scheme-theoretically to equal the persistent catalecticant/Lucas union; this is a contained-component theorem, not a consequence of Theorem 8.1. Second, effective base selection must use a single nonzero bad polynomial whose zero locus contains the entire union of Σ , both rulings, and the vertical/Chow loci. The report bounds each individual generator by degree four in each selected root, but does not show that one polynomial with the same bound contains the union. Multiplying component equations changes the stated threshold. These are explicit proof gates.

Proposition 8.3 (Conditional effective Hessian corollary). Assume the two gates of Remark 8.2, with a global bad polynomial of individual degree at most four. For syndrome degree $n \geq 5$, outside the persistent/Lucas union an integral reduced slice exists whenever

$$q \geq \min \left\{ \frac{(n-4)(n+3)}{2} + 1, 5(n-4) \right\}.$$

Its deletion degree is at most $3n - 4$. If also $q + 1 - 2\sqrt{q} > 3n - 4$, every split-free syndrome lies in that union.

Proof. For a split base R of degree $n-4$, the root-compatible Plücker coordinates are quadratic in its coefficients. The assumed global bad polynomial has individual root degree at most four. Schwartz–Zippel after adjoining the Vandermonde gives the first bound; evaluation on pairwise disjoint five-element blocks gives the second. The moving diagonal, determinant, branch, fixed-root collision, and final collision divisors have degrees

$$n-4, 2, 2, 2(n-4), 4,$$

whose sum is $3n-4$. The normalized integral $(2, 2)$ slice has genus at most one, so the point bound leaves a rational ordered residual root outside every deletion, and contraction lifts it squarefreely. $\square$

Theorem 8.1 is a containment theorem. It does not assert that every Lucas-carrier point is deep; that remains an arithmetic question for each level.

9 Lucas carriers and their arithmetic

Let $d = 2^s$ with $s \geq 2$ in characteristic two. Lucas’ theorem gives the top nucleus of the degree- d normal rational curve as

$$\mathbb{P}\langle e_1, \dots, e_{d-1} \rangle,$$

whose coherent degree- $(d+1)$ lift is

$$\mathcal{M}_{d+1} = \mathbb{P}\langle e_2, \dots, e_{d-1} \rangle.$$

At the distinguished Borel endpoint e_{d-1} the Hankel kernel contains

$$\mathcal{U}_s = \langle 1, t, t^d \rangle.$$

Proposition 9.1 (Lucas overlap kernel). For $d = 2^s$, the lower nucleus and its coherent lift are exactly

$$\mathbb{P}\langle e_1, \dots, e_{d-1} \rangle, \quad \mathbb{P}\langle e_2, \dots, e_{d-1} \rangle.$$

At e_{d-1} , $\mathcal{U}_s$ is endpoint-specific; the intersection common to the full carrier is only $\langle 1, t^d \rangle$.

Proof. Lucas’ theorem gives $\binom{d}{i} \equiv 0 \pmod{2}$ for $0 < i < d$. Applying this support condition to the two consecutive contraction rows removes the two endpoint coordinates and gives the displayed lift. For a basis point e_i , the Hankel equations kill the adjacent coefficients b_{i-1}, b_i . At $i = d-1$, the monomials $1, t, t^d$ survive. Varying i over the carrier leaves only $1, t^d$ in the intersection. $\square$

Proposition 9.2 (Linearized root cover). On the generic locus $BC \neq 0$, the normalized ordered-root cover of $A + Bt + Ct^d$ has geometric monodromy $\mathrm{AGL}_1(\mathbb{F}_d)$, minimal constant field $\mathbb{F}_d$, and based deck group C_{d-1} . Coefficientwise Frobenius acts on the based group by multiplication by two, with exact order s . The net contains a split squarefree member over $\mathbb{F}_{2^m}$ if and only if $s \mid m$.

Proof. After dividing by C , choose

$$\beta^{d-1} = B/C, \quad y^d + y = A/(C\beta^d).$$

The roots are $\{\beta(y+c) : c \in \mathbb{F}_d\}$. Their affine symmetries give $\mathrm{AGL}_1(\mathbb{F}_d)$. The Kummer extension has degree $d-1$ because the valuation of B/C at its simple zero is one. After adjoining β , the right side of the Artin–Schreier equation has a simple pole in A/C ; an

element of the form $z^d + z$ has pole order divisible by d . Hence the additive extension has degree d , the cover is connected, and its geometric monodromy is the full affine group.

Ratios of nonzero root differences recover $\mathbb{F}_d$, so this is the minimal constant field. Boundary members with $C = 0$ have degree at most one, while those with $B = 0$ have zero derivative; neither is split squarefree of degree d . Thus a split member forces $\mathbb{F}_d \subset \mathbb{F}_{2^m}$, equivalently $s \mid m$. Conversely $t^d + t$ splits simply under that inclusion. The equality $\text{ord}_{2^s-1}(2) = s$ follows from

$$\gcd(2^r - 1, 2^s - 1) = 2^{\gcd(r,s)} - 1.$$

□

At $s = 2$ this recovers the odd/even extension law at redundancies six and seven. The first fresh carrier is $s = 3$, in syndrome degree nine. The $\mathbb{F}_8$ -linear section has order-three component Frobenius, but its four quotient directions change the full answer.

Proposition 9.3 (The e_7 orbit). The stabilizer of $e_7 \in \mathbb{P}(\Gamma^9 E)$ is the upper Borel, so its PGL_2 -orbit is $\mathbb{P}^1$. At the distinguished point,

$$W_{e_7} = \langle 1, t, t^2, t^3, t^4, t^5, t^8 \rangle.$$

Proof. Lucas parity and extraction of the $x^2 y^7$ coefficient give

$$ge_7 = (ad + bc)^2 (b^5 e_2 + b^4 d e_3 + b d^4 e_6 + d^5 e_7).$$

This is proportional to e_7 exactly when $b = 0$. The apolar action is contragredient, so it transports both W_f and split root divisors. The displayed kernel follows from the two Hankel equations. □

Proposition 9.4 (Framed Artin–Schreier quotient). Order roots $0, 1, a, b, c, d, u, v$ and put

$$S = a + b + c + d, \quad E = ab + ac + ad + bc + bd + cd, \quad h = 1 + S, \quad p = 1 + E + S + S^2.$$

On the open set where the six fixed roots are distinct, $h, p, 1 + h + p$ are nonzero, and $x^2 + hx + p \neq 0$ for $x = a, b, c, d$, the remaining pair is the geometrically integral Artin–Schreier cover

$$y^2 + y = p/h^2.$$

Its rational lifting law is $\text{Tr}_{\mathbb{F}_{2^m}/\mathbb{F}_2}(p/h^2) = 0$.

Proof. The equations $e_1 = e_2 = 0$ give $u + v = h$ and $uv = p$. On $h \neq 0$, put $y = u/h$. Along $h = 0$, the leading coefficient of p/h^2 is

$$D = 1 + bc + bd + cd + B + B^2,$$

and $\partial D / \partial b = 1 + c + d \neq 0$, so D is not a square in the residue field. If $g^2 + g$ has a pole of order two, its leading coefficient is the square of the principal coefficient of g . Therefore the Artin–Schreier class remains nonzero after extending constants, proving geometric integrality. The trace criterion gives the rational lifting law. □

Proposition 9.5 (Additive frame cover). The family

$$t^8 + A_4 t^4 + A_2 t^2 + A_1 t + A_0, \quad A_1 \neq 0,$$

has a connected affine-frame cover with geometric monodromy $\text{AGL}_3(\mathbb{F}_2)$.

Proof. For an $\mathbb{F}_2$ -space U and $w \notin U$,

$$L_{U+\langle w \rangle} = L_U^2 + L_U(w)L_U.$$

Induction gives exponents 8, 4, 2, 1 and $A_1 = \prod_{v \neq 0} v \neq 0$. An affine frame is an origin plus three independent differences. The coefficient map is generically finite with exactly $8|\mathrm{GL}_3(\mathbb{F}_2)| = 1344$ frames in a fiber. Its frame function field therefore has degree 1344 over the coefficient field and invariant field equal to that coefficient field. $\square$

Theorem 9.6 (Shalowness of the degree-nine e_7 orbit). For every $m \geq 3$, the entire PGL_2 -orbit of e_7 is shallow over $\mathbb{F}_{2^m}$. The number of monic additive-affine witnesses is

$$\frac{q(q-1)(q-2)(q-4)}{1344}.$$

Proof. Every three-dimensional $\mathbb{F}_2$ -subspace $V \subset \mathbb{F}_{2^m}$ gives

$$L_V(t) = \prod_{v \in V} (t - v) = t^8 + A_4 t^4 + A_2 t^2 + A_1 t.$$

The t^7, t^6 coefficients vanish, so $L_V \in W_{e_7}$, and $A_1 \neq 0$ makes its roots simple. Every admissible binary redundancy-ten field has $m \geq 4$. Proposition 9.3 transports the witness around the orbit.

There are

$$\binom{m}{3}_2 = \frac{(q-1)(q-2)(q-4)}{168}$$

three-spaces, each with $q/8$ affine translates. Since $L_{a+V}(t) = L_V(t) + L_V(a)$ and the affine root set recovers its difference space and coset, there is no overcount. $\square$

Thus the order-three law of Proposition 9.2 belongs to a proper $\mathbb{F}_8$ -linear section, not to deepness of the full e_7 kernel. The other intrinsic strata of the six-dimensional degree-nine carrier are not classified here.

10 Verification and formal boundary

The conceptual theorems above are accompanied by deterministic, committed evidence bundles. Table 7 records which claims use exhaustive computation. Every named bundle contains the primary generator, a compact JSON or text certificate, an independent replay where stated, and a SHA-256 manifest. The load-bearing generator, certificate, and replay hashes were rechecked during manuscript integration. The C491 and C498 manifests also include their living theorem reports; those two report-only hashes became stale after later closeout expansions. No computational artifact hash is stale, and no predecessor manifest is silently rewritten here. The public names Certificate R5 and Certificate R6 refer to those bundles through the release-level schema; the C-numbers are retained only as provenance.

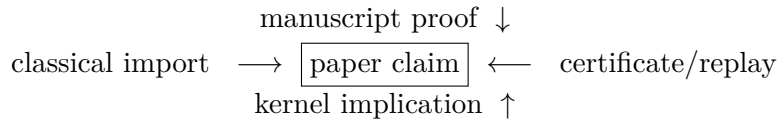


Figure 4: Claim-level trust routes. A mixed theorem may use several routes, but each route keeps its own boundary: certificates close finite domains, imports retain their hypotheses, and Lean proves only the printed implication.

Table 7: Public verification map. Exact filenames, commands, hashes, domains, and stop conditions belong to the electronic supplement.

Public label	Verified domain	Replay and trust boundary
Certificate R5	nineteen finite fields	independent pointwise and orbitwise replay; canonical representatives and exhaustion records
Certificate R6	direct scan through $q = 16$; finite bridge below 29	independent syndrome replay and separate radius gate
Certificate R6-NF	small exceptional normal forms	deterministic normal-form reconstruction; not a second implementation
Certificate R7	$q = 7, 8, 9, 11$ and finite bridge below 37	independent five-secant/orbit checks; split-free and radius flags remain separate
Certificate R8	fixed-level characteristic cases and bounds	independent algebra/nucleus replay; not an ambient census
Certificate R9	residual algebra and transported slices	independent residual/slice replay; geometric integrality remains a paper proof
Certificate R9-49	characteristic-seven carrier at $q = 49$	one exhaustive carrier implementation; not a whole-code census
Certificate Hessian	bounded ordered-Hessian algebra	regression only; does not replace the geometric component proof
Certificate Lucas	recorded Lucas parameter domain	independent arithmetic replay
Certificate e7	quotient-cover open set and additive specialization	independent quotient-cover replay

The public labels are defined by `supplement/CERTIFICATE-SCHEMA.md`. Literal development commands and working directories are recorded in `supplement/REPRODUCING.md` and `verification-map.md`. The release manifest is intentionally marked incomplete until it contains a repository URL, immutable tag and commit, archive identifier, DOI, byte counts, and hashes. Consequently the present development draft is not an externally archived proof-complete release.

The redundancy-nine residual algebra and synthesis implication are also formalized in Lean. The import gate and its axiom-audit module are

```
RelativeConicArcs.Gates.PRSRedundancyNine,
RelativeConicArcs.Gates.PRSRedundancyNineAxiomAudit.
```

Lean checks divided-power contraction, both residual Hankel identities, the determinant and branch formulas, the residual roots and their sum/product, the witness-to-shallow implication, the persistent-family cardinality, and the orbit-count table from explicit arithmetic-case and family-count inputs. It does not derive that table from a formal $\mathrm{PGL}_2/\mathrm{PTL}_2$ action. Geometric integrality, Hasse–Weil existence after deletions, identification of geometric families with PRS syndromes, and exhaustion remain explicit structure fields rather than hidden axioms. The audit reports only the standard `propext`, `Classical.choice`, and `Quot.sound` dependencies.

11 Provenance and responsibility

This research was human-directed and agent-generated. The author directed the programme and takes responsibility for the manuscript; AI systems are not authors. Section 10 assigns

Table 8: Separation of mathematical input from kernel-checked formalization.

Result	Mathematical input	Kernel-checked content
R9 residual synthesis	six-slice component theorem, genus/deletion bounds, coding identification, and exhaustion	divided-power contraction, residual identities, determinant/branch formulas, roots, and witness-to-shallow implication
R9 family inventory	the actual $\mathrm{PGL}_2/\mathrm{PTL}_2$ action and geometric family identification	arithmetic case inputs, family-count inputs, and the resulting numerical table

every published claim an exact trust route. The corresponding source, certificates, replays, and pinned Lean targets will be archived at the paper-specific export repository and versioned DOI recorded in the release manifest; the development monorepo is not a publication artifact. The specific model and tool disclosure will follow the policy of the selected repository and journal. No placeholder archive field is presented here as a completed public release.

12 Exact boundary and further questions

The results proved here do not constitute an arbitrary-redundancy classification.

1. Redundancy five is complete for every $q \geq 7$.
2. Redundancy six is a complete all-field deep-hole classification. Redundancy seven has a complete all-field split-free classification; its $q = 7, 8, 9$ covering-radius gate remains separate.
3. Redundancy eight is complete for $q \geq 43$; no bounded-field completion is inferred. Redundancy nine has a complete transverse argument for $q \geq 53$ but retains its slice/base-selection and residual component gates.
4. The characteristic-two ordered-Hessian theorem classifies the geometric degeneracy strata. Its root-compatible pullback and effective base-selection constant retain the stated gates, and it does not classify arithmetic deepness on every Lucas carrier.
5. The e_7 orbit is only one intrinsic stratum of the degree-nine Lucas carrier. The other strata, and therefore a redundancy-ten synthesis, remain open.
6. The effective polar-induction theorem requires a new level-specific monodromy/component calculation at each redundancy. A field census cannot replace this input.

The structural conclusion is nevertheless stable. Beyond redundancy four, deepness is no longer controlled by one invariant or by isolated bad fibers. The persistent geometry is a coherent rank-two polar flag; the modular geometry is a Lucas-computed contraction kernel; and the remaining arithmetic is the Frobenius action on a normalized ordered-root cover. Redundancy five is the first exceptional-cover case, while the later theorems show how that cover is propagated, contained, or replaced in the levels and carrier strata treated here.

A Statement-adequacy ledger

This appendix reproduces verbatim the formal statements on which the manuscript places Lean-level trust. It is intended to make statement adequacy refereeable without requiring the reader to inspect the full development. The source anchors are `RelativeConicArcs/`

PRSResidualQuadratic.lean and RelativeConicArcs/PRSRedundancyNine.lean; the import and audit gates are those named in Section 10.

Bottom definitions.

```
def dividedPowerContraction {n : ℕ} (r : R)
  (a : Fin (n + 2) → R) : Fin (n + 1) → R :=
  fun i => a i.succ - r * a i.castSucc

structure ResidualSliceInput (S : Type*) (q : ℕ) where
  isDeep : S → Prop
  exceptional : S → Prop
  integralGenusAtMostOneSlice : S → Prop
  rationalPointOutsideDeletedDivisors :
    q ≥ 53 → S → Prop
  splitSquarefreeWitness : S → Prop
  witnessMakesShallow :
    ∀ {s}, splitSquarefreeWitness s → ¬ isDeep s
  pointGivesWitness :
    ∀ (hq : q ≥ 53) {s}, exceptional s →
      integralGenusAtMostOneSlice s →
        rationalPointOutsideDeletedDivisors hq s →
          splitSquarefreeWitness s

structure PersistentFamilyData (S : Type*) (q : ℕ)
  [Fintype S] [DecidableEq S] where
  tangent : Finset S
  sigma : Finset S
  deep : Finset S
  tangent_sigma_disjoint : Disjoint tangent sigma
  deep_eq : deep = tangent ∪ sigma
  fieldOrderPositive : 0 < q
  tangent_card : tangent.card = q * (q + 1)
  sigma_card_doubled : 2 * sigma.card = q * (q ^ 2 - 1)
  orbitCase : OrbitArithmeticCase
```

The first definition is the paper’s contraction. The two structures make geometric integrality, deleted-divisor rational points, witness production, family identification, and family counts explicit inputs rather than axioms hidden inside a Boolean certificate.

Headline synthesis terminal.

```
theorem redundancyNineSynthesis {S : Type*} {q : ℕ}
  [Fintype S] [DecidableEq S]
  (hq : q ≥ 53)
  (slice : ResidualSliceInput S q)
  (families : PersistentFamilyData S q)
  (exceptional_iff :
    ∀ s, slice.exceptional s ↔ s ∉ families.deep)
  (persistentDeep :
    ∀ s, s ∈ families.deep → slice.isDeep s)
  (hslice :
    ∀ s, slice.exceptional s →
      slice.integralGenusAtMostOneSlice s)
  (hpoint :
    ∀ s (hs : slice.exceptional s),
      slice.rationalPointOutsideDeletedDivisors hq s) :
  (∀ s, slice.isDeep s ↔ s ∈ families.deep) ∧
  (∀ s, s ∉ families.deep →
    slice.splitSquarefreeWitness s) ∧
  families.deep.card = q * (q + 1) ^ 2 / 2 ∧
  (projectiveOrbitCount families.orbitCase,
```

```

semilinearOrbitCount families.orbitCase) =
orbitCountPair families.orbitCase

```

This terminal is adequate for the final synthesis implication once the paper supplies the arguments `exceptional_iff`, `persistentDeep`, `hslice`, and `hpoint`. It does not prove those geometric and coding inputs, nor does it derive the orbit table from a formal group action.

Axiom audit. The guarded audit of fourteen terminals reports no axioms for the finite witness and orbit-pair terminals; only `propext` and `Quot.sound` for ring normalization; and only `propext`, `Classical.choice`, and `Quot.sound` for field/root terminals, the family cardinality, and the headline synthesis theorem. It uses no project-specific axiom, native evaluator, finite-data import, or opaque computational oracle.

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